

Generation Interconnection System Impact Study Report

For Queue Project Q22-1990

Harrisburg Facility 138 kV

December 04, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1990. The Interconnection Customer proposes a 111.0 MW battery energy storage facility interconnecting at the Harrisburg Facility 138 kV substation in the Pennsylvania area. The request was received on January 23, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 506 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,250,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ayer / Talenenergy Martins Crk Llc Harrisburg 138.0 kV
Contingency: Talenenergy Martins Crk Llc Harrisburg / Little Chute 345.0 kV L
Loading: 140.5% (Rating: 506 MVA)
DFAX: 21.2%
Pre-project loading: 135.8%
2. Facility: Pingston / Jack Crk 138.0 kV L
Contingency: Little Chute Substation / Talenenergy Martins Crk Llc Harrisburg 345.0 kV
Loading: 140.3% (Rating: 242 MVA)
DFAX: 8.2%
Pre-project loading: 136.6%
3. Facility: Jack Creek / Lakeland Solar Energy Llc 138.0 kV line
Contingency: Little Chute / Talenenergy Martins Crk Llc Harrisburg 345.0 kV
Loading: 137.7% (Rating: 210 MVA)
DFAX: 6.3%
Pre-project loading: 134.4%
4. Facility: Ayer / Paxton Crk Cogeneration 138.0 kV
Contingency: Talenenergy Martins Crk Llc Harrisburg / Little Chute 345.0 kV
Loading: 137.6% (Rating: 537 MVA)
DFAX: 21.2%
Pre-project loading: 133.2%
5. Facility: Panguitch / Elk City Llc Substation 138.0 kV Line
Contingency: Little Chute / Talenenergy Martins Crk Llc Hrrisburg 345.0 kV
Loading: 137.1% (Rating: 302 MVA)
DFAX: 8.2%
Pre-project loading: 134.1%
6. Facility: Calaveras / Panguitch 138.0 kV
Contingency: Little Chute / Talenenergy Martins Creek Llc Harrisburg 345.0 kV
Loading: 137.0% (Rating: 287 MVA)

DFAX: 8.2%

Pre-project loading: 133.8%

7. Facility: Dumbarton Tap / Paxton Crk Cogeneration 138.0 kV

Contingency: Little Chute / Talenenergy Martins Crk Llc Harrisburg 345.0 kV

Loading: 136.2% (Rating: 554 MVA)

DFAX: 21.2%

Pre-project loading: 131.9%

8. Facility: Pingston / Calaveras 138.0 kV

Contingency: Little Chute / Talenenergy Martins Creek Llc Harrisburg 345.0 kV

Loading: 135.0% (Rating: 271 MVA)

DFAX: 8.2%

Pre-project loading: 131.7%

9. Facility: Dumbarton Tap / Macathur 138.0 kV

Contingency: Little Chute / Talenenergy Martins Crk Llc Harrisburg 345.0 kV L

Loading: 132.8% (Rating: 567 MVA)

DFAX: 16.6%

Pre-project loading: 129.5%

10. Facility: Macarthur / Ft Gage Substation 138.0 kV L

Contingency: Taenenergy Martins Creek Llc Harrisburg / Little Chute 345.0 kV

Loading: 131.4% (Rating: 586 MVA)

DFAX: 16.6%

Pre-project loading: 128.2%

11. Facility: 12Th & Irving / Johnhart 138.0 kV Line

Contingency: Little Chute / Talenenergy Martins Creek Llc Harrisburg 345.0 kV

Loading: 126.7% (Rating: 252 MVA)

DFAX: 7.5%

Pre-project loading: 123.4%

12. Facility: Ft Gage / Hunt 138.0 kV

Contingency: Talenenergy Martins Crk Llc Harrisburg / Little Chute 345.0 kV L

Loading: 119.0% (Rating: 340 MVA)

DFAX: 10.8%

Pre-project loading: 115.5%

13. Facility: Little Chute / Hunt 138.0 kV

Contingency: Little Chute Substation / Talenenergy Martins Creek Llc Harrisburg 345.0 kV Line

Loading: 115.5% (Rating: 364 MVA)

DFAX: 10.8%

Pre-project loading: 112.2%

14. Facility: 12Th & Irving / Lakeland Solar Energy Llc 138.0 kV

Contingency: Little Chute / Talenenergy Martins Crk Llc Harrisburg Substation 345.0 kV

Loading: 115.4% (Rating: 269 MVA)

DFAX: 6.3%

Pre-project loading: 112.8%

15. Facility: Talenenergy Martins Creek Llc W Shore / Broussard 138.0 kV Line

Contingency: Dumbarton Tap / Paxton Crk Cogeneration 138.0 kV

Loading: 111.7% (Rating: 238 MVA)

DFAX: 41.1%

Pre-project loading: 92.5%

16. Facility: Paxon / L L Wilkins 138.0 kV

Contingency: Paxon / S Lake Geneva 138.0 kV

Loading: 109.9% (Rating: 280 MVA)

DFAX: 13.9%

Pre-project loading: 104.3%

17. Facility: Garland / Usaa 2 138.0 kV

Contingency: Macarthur / Ft Gage 138.0 kV

Loading: 109.0% (Rating: 251 MVA)

DFAX: 7.7%

Pre-project loading: 105.6%

18. Facility: Usaa 2 / Coronet 138.0 kV

Contingency: Ft Gage / Austin Dam 138.0 kV

Loading: 108.0% (Rating: 377 MVA)

DFAX: 9.5%

Pre-project loading: 105.2%

19. Facility: Orange City / Broussard Substation 138.0 kV

Contingency: Dumbarton Tap / Paxton Creek Cogeneration 138.0 kV

Loading: 106.9% (Rating: 216 MVA)

DFAX: 41.1%

Pre-project loading: 85.7%

20. Facility: Pedley / Bathurst 138.0 kV

Contingency: Ft Gage / Macarthur 138.0 kV L

Loading: 106.6% (Rating: 183 MVA)

DFAX: 9.3%

Pre-project loading: 101.0%

21. Facility: Talenenergy Martins Creek Llc Harrisburg Substation 345/138 kV Auto

Contingency: Talenenergy Martins Crk Llc Harrisburg 500-345 kV Transformer

Loading: 105.2% (Rating: 477 MVA)

DFAX: 22.0%

Pre-project loading: 100.0%

22. Facility: Allegheny Hydro No 8 / Eklutna 345.0 kV

Contingency: Piney Creek Project / Caldwell 345.0 kV line

Loading: 104.3% (Rating: 1,188 MVA)

DFAX: 5.2%

Pre-project loading: 103.8%

23. Facility: Wolverine Crk / Yorkshire 138.0 kV

Contingency: South Lake Geneva / Paxon 138.0 kV Line

Loading: 103.6% (Rating: 266 MVA)

DFAX: 13.9%

Pre-project loading: 97.8%

24. Facility: Wolverine Creek / L L Wilkins 138.0 kV

Contingency: Paxon / S Lake Geneva 138.0 kV

Loading: 102.4% (Rating: 287 MVA)

DFAX: 13.9%

Pre-project loading: 97.0%

25. Facility: Fyette Substation / Mercury 345.0 kV

Contingency: Bixby Hill / Brunner Island 500.0 kV Line

Loading: 101.0% (Rating: 1,190 MVA)

DFAX: 7.4%

Pre-project loading: 100.3%

26. Facility: 12Th & Irving Substation / Bathurst 138.0 kV L

Contingency: Macarthur / Ft Gage 138.0 kV

Loading: 100.6% (Rating: 221 MVA)

DFAX: 9.3%

Pre-project loading: 95.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ayer / Talenenergy Martins Crk Llc Harrisburg Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$604,000	17% (\$103,000)
Pingston / Jack Creek 138.0 kV	Rebuild line with higher-capacity conductor	\$6,148,000	100% (\$6,148,000)
Jack Creek / Lakeland Solar Energy Llc 138.0 kV line	Rebuild line with higher-capacity conductor	\$1,209,000	22% (\$268,000)
Ayer / Paxton Crk Cogeneration 138.0 kV	Rebuild line with higher-capacity conductor	\$4,988,000	17% (\$851,000)
Panguitch / Elk City Llc Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,938,000	100% (\$1,938,000)
Calaveras / Panguitch 138.0 kV	Rebuild line with higher-capacity conductor	\$2,341,000	100% (\$2,341,000)
Dumbarton Tap / Paxton Crk Cogeneration 138.0 kV	Rebuild line with higher-capacity conductor	\$1,991,000	17% (\$340,000)
Pingston / Calaveras Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$5,108,000	100% (\$5,108,000)
Dumbarton Tap / Macathur 138.0 kV	Rebuild line with higher-capacity conductor	\$1,218,000	10% (\$125,000)
Macarthur / Ft Gage Substation 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,550,000	10% (\$159,000)
12Th & Irving / Johnhart 138.0 kV Line	Rebuild line with higher-capacity conductor	\$11,721,000	26% (\$2,999,000)
Hunt / Fortgage 138.0 kV	Rebuild line with higher-capacity conductor	\$1,332,000	12% (\$166,000)
Little Chute / Hunt 138.0 kV	Rebuild line with higher-capacity conductor	\$3,034,000	12% (\$379,000)
Lakeland Solar Energy Llc / 12Th & Irving 138.0 kV	Rebuild line with higher-capacity conductor	\$2,210,000	22% (\$490,000)
Talenenergy Martins Creek Llc W Shore / Broussard 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,163,000	68% (\$2,134,000)
Paxon / L L Wilkins 138.0 kV	Reconductor line	\$605,000	34% (\$203,000)
Garland / Usaa 2 138.0 kV	Reconductor line	\$2,437,000	16% (\$382,000)
Usaa 2 / Coronet 138.0 kV	Reconductor line	\$2,140,000	8% (\$171,000)
Orange City / Broussard 138.0 kV L	Reconductor line	\$1,065,000	68% (\$718,000)
Pedley / Bathurst 138.0 kV	Reconductor line	\$336,000	100% (\$336,000)
Talenenergy Martins Creek Llc Harrisburg 345/138kV Transformer Bank	Replace transformer with higher-rated unit	\$14,794,000	40% (\$5,980,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Allegheny Hydro No 8 / Eklutna 345.0 kV	Reconductor line	\$31,330,000	3% (\$907,000)
Wolverine Crk / Yorkshire 138.0 kV	Reconductor line	\$1,179,000	34% (\$396,000)
Wolverine Crk / L L Wilkins 138.0 kV	Reconductor line	\$754,000	34% (\$253,000)
Fayette Substation / Mercury 345.0 kV	Reconductor line	\$8,574,000	2% (\$138,000)
12Th & Irving Substation / Bathurst 138.0 kV	Reconductor line	\$1,217,000	100% (\$1,217,000)
	Total allocated to project		\$34,250,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.